INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

JUNEJA STREET

Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

Vihaan Juneja | Grade 9

Strategy development and research

Aarav Juneja | Grade 11

Risk analysis and execution
github.com/V-Juneja/ml-supertrend

Investment hypothesis and complete strategy

The strategy seeks short-term rebounds after unusually strong selling in volatile stocks. A fresh low alone is not a BUY: price must confirm an upward Supertrend reversal, recent oversold RSI and signal spacing. The machine-learning layer learns online from completed hypothetical long and short probes. Three model components use rolling outcomes, adverse-movement-weighted batches and an 8 by 8 market-context memory to adjust the Supertrend multiplier and ATR smoothing within bounds. The learning changes signal sensitivity and timing; explicit portfolio rules still set entry, position size and exit. The hypothesis fails when selling reflects lasting information, or costs erase any rebound.

Each probe opens at a candle open and matures at the close of its fifth candle. A move of one entry ATR gives the long probe a score of +1 and the matching short probe -1. Only matured outcomes can update later settings. Because probes overlap, their labels are dependent and are not account returns.

At equal confidence, one ATR of adverse movement halves a probe's batch weight. Context-memory influence rises with its supporting observations but is capped at 65%. A bounded update layer smooths parameter proposals; portfolio rules separately set entries, size and exits.

Why these stocks

The scan uses liquid S&P 500 stocks so spreads and trading capacity are less likely to overwhelm a short-horizon signal. DELL is shown because its larger price swings make the reversal bands and signal timing easy to inspect. Portfolio statistics below summarize the saved ranked BUY lists across the universe. Each BUY must pass liquidity and portfolio risk checks before sizing.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for liquid S&P 500 stocks. BUY opens one long; repeated BUYs do not add size. A confirmed SELL or protective stop closes it. Historical replay enters at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$
$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the multiplier. The boundaries sit mA above and below x; the baseline uses l = 24 and m = 1.4. They trail or reset around price. A close across the previous opposing band flips trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

The rolling learner uses up to 30 matured probes per direction: below 38% wins with mean score < 0 it proposes wider bands and longer ATR; above 62% with mean score > 0 it proposes narrower, faster settings. The batch learner weights each probe by confidence / (1 + adverse excursion in ATR units), then summarizes normalized gain, profit factor and downside. A 1-ATR adverse move halves its weight.

Context memory conditions probe outcomes on market state. Its 0–1 trend score combines 35% inverse 64-candle entropy, 35% scaled 64-candle Hurst persistence and 30% normalized ADX(14); 0 is choppy, 1 trending. Relative volatility is current ATR / trailing 100-candle mean, scaled 0–1 at 2.5×. The two axes index 8×8 = 64 cells. Each cell exponentially smooths probe gain, downside and parameter proposals (50-update half-life). A Gaussian-weighted 3×3 neighborhood forms its local proposal; sample count sets the local/global blend, capped at 65%. This adapts indicator settings, not capital allocation.

The combiner blends rolling and context proposals with available batch feedback. Larger multiplier requires a larger reversal; longer ATR responds more slowly. Deadbands, update caps, parameter bounds and baseline reversion limit drift. Separate portfolio rules control capital.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current multiplier or ATR length; theta tilde is the rounded, bounded target; alpha is the update fraction. For multiplier 1.40, target 1.50 and alpha 0.20, the update is 1.42. Deadbands ignore tiny changes; caps constrain each update and periodic reversion pulls settings toward baseline. The multiplier stays 0.5-5.0 and ATR length 5-100.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within the prior 50 candles and at least 10 candles since the last signal. RSI compares Wilder-smoothed gains and losses over 14 candles on a 0–100 scale. SELL requires a fresh-high reversal, downward flip and recent RSI above 70. Signals use completed candles; volume and tick-pressure filters are off.

DELL check. 780 bars; 776 matured probes per side; 6 BUY / 8 SELL; multiplier 1.40–2.145; ATR 24. Prefix checks matched.

2 ML Supertrend replay and backtest charts

Archived replay: ML Supertrend BUY emails, 8 September 2025–14 October 2026; the last list carried through 30 October (290 sessions). Apply each list at the next session open, with $100,000 starting equity and 10 basis points per side. Backtest sizing allowed ten holdings; the declaration allows up to ten names at 10% each.

ML Supertrend replay result	Value
Ending marked equity	$151,542.68
Total return / annualized growth	+51.54% / +43.92%
Maximum drawdown	-21.10%
QQQ benchmark return	+19.39%

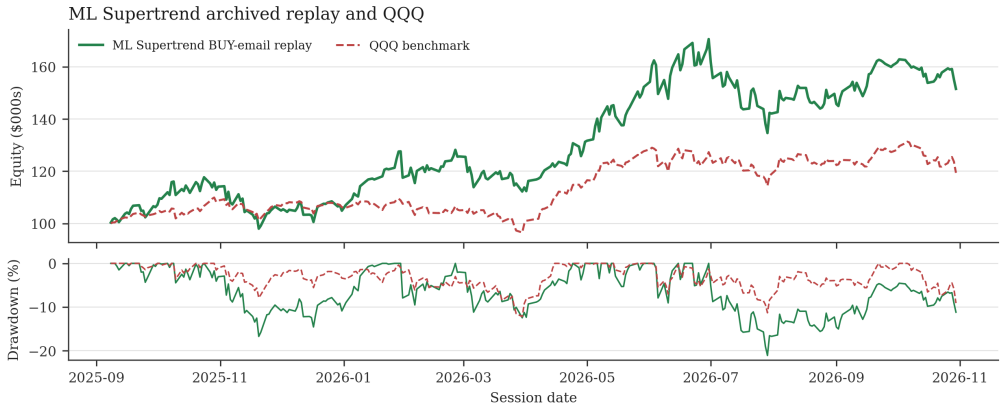


Figure 1. Ending equity was $151.5k versus QQQ at $119.4k; the replay reached a 21.10% maximum drawdown. Data: Alpaca SIP split-adjusted bars; our replay.

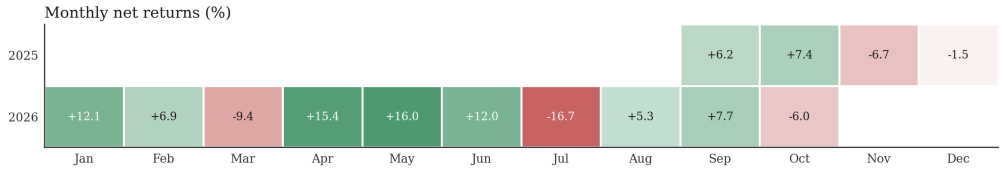


Figure 2. Nine of 14 months were positive; the largest monthly loss was 16.7% in July 2026. Monthly results inform position sizing and drawdown limits. Data: same replay.

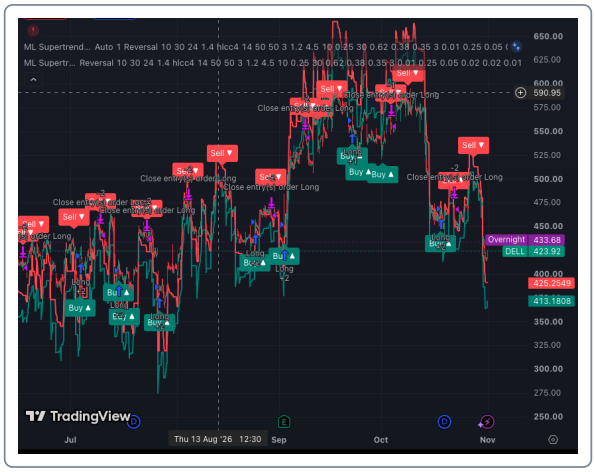


Figure 3. Team TradingView DELL 30-minute paper-trading chart, July–November 2026. A separate Yahoo Finance indicator replay on 780 bars through 25 September generated 6 BUYs, 8 SELLs and 776 completed probes per side. Portfolio results above come from the archived email replay.

3 Trade expectancy and risk tests

Net Trade Expectancy Equation

$$e = pW - (1-p)D, \qquad p_{\text{BE}} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

Trade statistic	Historical result
Completed trades	622
Win rate	46.62%
Mean net profit / return per trade	+$81.73 / +0.8173%
Median trade return	-0.6268%
Average net winning / losing return	+8.05% / -5.50%
Profit factor	1.28
Break-even win rate at observed payoffs	40.59%
Best / worst net trade return	+62.80% / -40.03%
Longest losing streak	15 trades

These 622 closed trades come from the ML Supertrend email replay. The 46.62% win rate exceeded its 40.59% break-even level; profit factor was 1.28. The median trade lost 0.6268%, while larger gains lifted the mean. Open positions are excluded. Trade calculations and fills: GitHub trade evidence.

ML Supertrend one-year Monte Carlo

We resampled sequential equity changes from the archived ML Supertrend replay (290 sessions, 8 September 2025–30 October 2026) in circular five-session blocks. Each of 50,000 paths spans 252 sessions. Source, calculation and chart: Monte Carlo files.

One-year path statistic	Result
Median terminal return	+43.98%
5th / 95th percentile terminal return	-11.85% / +131.43%
Paths ending below start	10.87%
Mean return in worst 5% of paths	-21.93%
95th percentile maximum drawdown	35.46%

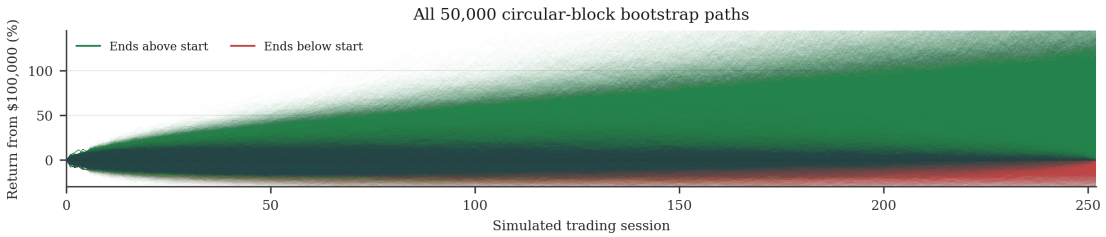


Figure 4. All 50,000 resampled paths are shown: green ended above the start; red below. Although the median one-year return was +43.98%, 10.87% lost money; the 5th-percentile return was -11.85%. These scenarios show a wide range of outcomes, not a single expected path.

4 Portfolio construction and risk

False reversals can cluster and technology names can fall together. Entry spacing, stop-risk budgets, concentration caps and account-loss rules address these strategy-specific risks. Sizing is tested at three fixed ticket sizes below. Risk-test assumptions and code.

Control	Rule and purpose
Exposure	Long-only, no borrowing; at most ten names, 10% of equity per name and 100% gross stocks. Ten full-size positions use the full allocation.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Trade log	Use one account; reject duplicate BUY entries. Record each signal, setting, order, fill, fee, stop and exit reason. Freeze the declared rules; close by the competition cutoff.

Position Sizing Equation

$$d_i = 2\,ATR_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.10E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

Fixed ticket / name	Avg. gross	Total return	Max drawdown	Sharpe (rf=0)
$5,000 (5%)	44.68%	+26.21%	-13.26%	1.13
$10,000 (10%)	80.30%	+51.54%	-21.10%	1.22
$15,000 (15%)	92.88%	+32.90%	-31.78%	0.77

Sizing replay: same saved BUY rankings, 10 bps per side, exits and dates (8 September 2025–30 October 2026), with fixed tickets based on $100,000 starting equity and at most ten names. Ten percent had the highest return and Sharpe of the three tested; 5% had the shallowest drawdown, while 15% had the deepest. Method · Results.

At a DELL price of $563.17, a $10,000 ticket allows 17 shares before the stop-risk check. A $500 planned-loss budget sets the maximum stop distance at $29.41 per share. The 10% ticket had the highest return and Sharpe among the three tested settings.

Final declared strategy

The strategy trades confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold up to ten names, capped at 10% each and 100% gross exposure. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.